Lecture 5

IT Project Time Management

1- Activity Definition

- identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables
- **Activity definition involves developing** a **more detailed WBS**
- **Methods**
  - **Activity Lists and Attributes**
    - **List that includes** the activities name, identifier or number, description
    - **Attributes** provide more information such as **predecessors, successors, logical relationships, resource requirements, constraints, imposed dates, and assumptions related to the activity**
  - **Milestones** a **significant event** that normally **has no duration**
  - **SMART Criteria**

2- Activity Sequencing

- identifying and documenting the relationships between project activities
- **Relationship types**
  - **Finish-to-Start (FS) -> one cannot start before other**
  - **Start-to-Start (SS) -> both can start at the same time**
  - **Finish-to-Finish (FF) -> one cannot finish unless other is complete**
    - **Example: Testing cannot finish** until **coding is complete**
  - **Start-to-Finish (SF) -> one cannot start until other finishes**
    - **Example:** The **old version remains active until the new version starts running**

- **Network Diagram (1)**
  - Focuses on **task dependencies and relationships** between activities
  - **Types**
    1. **Arrow Diagramming Method (ADM) or Activity-on-Arrow (AOA)**
       - **arrows** to represent activities and **nodes** for events
       - Only **Finish-to-Start (FS)** relationships are possible.
       - **Example:** X-> Y ->Z

    2. **Precedence Diagramming Method (PDM)**
       - **Arrows** to represent **relationships and boxes for activities**
       - shows **different types of dependencies.**
- **Gantt Charter (2)**
  - lists project activities and **their corresponding start and finish dates** in a calendar format
  - **Supports all types of task dependencies**
  - **Symbols**
    1. **A black diamond: a milestones**
    2. **Thick black bars: summary tasks**

3. **Lighter horizontal bars: durations of tasks**
4. **Arrows: dependencies between tasks**
5. **Uncolored diamond: a slipped milestone (delayed past its planned date)**

3- Activity Resource Estimation

- estimating how many resources a project team should use to perform project activities

4- Activity Duration Estimating

- estimating the number of work periods that are needed to complete individual activities
- **Three-Point Estimate** (instead of estimating only one number estimate 3)
  - Optimistic, most likely, and pessimistic
  - **Example:** three weeks for the optimistic, four weeks for the most likely, and five weeks for the pessimistic estimate
- **Program Evaluation and Review Technique (PERT)**
  - network analysis technique used to estimate project duration when there is a **high degree of uncertainty**
  - **Formula:** (optimistic time + 4X most likely time + pessimistic time) / 6

5- Schedule Development

- compiling everything together by analyzing
  a. activity sequences,
  b. activity resource estimates,
  c. activity duration estimates to **create the project schedule**

6- Schedule Control

- controlling and managing changes to the project schedule

Time Management Best Practices

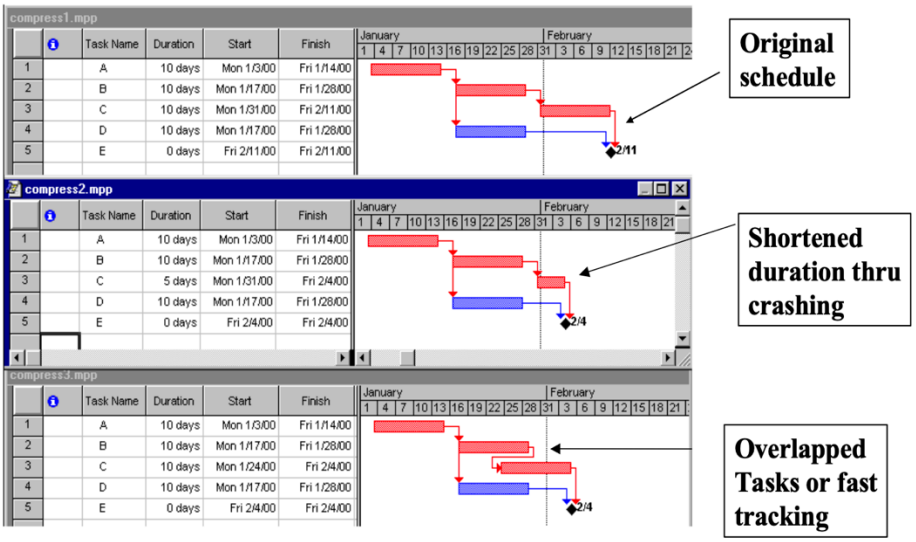
- **Critical Path Method (CPM) ->** network diagramming technique used **to predict total project duration**

- Critical path is the ***longest path*** through the **network diagram** and has the **least amount of slack or float (time an activity may be delayed without delaying a succeeding activity or the project finish date)**

- **Calculating the Critical Path**
  1. Create a Network Diagram– Show task dependencies.
  2. Estimate Task Durations– Assign time for each task.
  3. Identify All Possible Paths– Trace all task sequences.
  4. Find the Longest Path– This is the **Critical Path**.
  5. More Than One Critical Path?– If multiple paths have the same longest duration, they are all critical.
  6. Delays on the Critical Path?– The project **will be delayed** unless managed.
  7. Critical Path Changes Over Time– Updates may shift the path as tasks progress.

Techniques for Shortening a Project Schedule

- **Shorten durations** of **critical tasks by adding more resources or changing their scope**
- **Accepting delays**
- **Crashing tasks** by **obtaining the greatest amount of schedule compression** for **the least incremental cost**
- **Fast tracking tasks by doing them in parallel** or overlapping them



Buffers and Critical Chain

- **Buffer** is extra time added to complete a task.
- **Murphy's Law**
  - **"If something can go wrong, it will."**
  - **Solution:** Plan for delays (like PERT pessimistic time).
- **Parkinson's Law**
  - **"Work expands to fill the time available."**
  - **Solution:** Set shorter deadlines & monitor progress.

- **Critical Chain Scheduling (CCS) -- avoiding wasted time.**
  - **Removes task-level buffers** and **instead adds**:
    - **Project Buffer**– Extra time before project deadline.
    - **Feeding Buffers**– Extra time before critical path tasks.

  - **Goal:** Reduce multitasking & improve efficiency.